

MATH 3312, SPRING 2026: HOMEWORK 7

Definition 1. A **principal ideal domain** or **PID** is an integral domain in which every ideal is principal.

- (1) Decide if the following statements are true or false. Give a line or two in justification.
 - (a) The quotient ring $k[x]/I$ with $I \leq k[x]$ an ideal is a field if and only if $I = (q(x))$, where $q(x) \in k[x]$ is an irreducible polynomial.
 - (b) There exists an injective k -linear transformation $T : V \rightarrow W$ between two k -vector spaces of dimension 4 that is *not* an isomorphism.
 - (c) If V is a k -vector space admitting a k -subspace W such that $W \simeq k^2$, and $V/W \simeq k^3$, then $V \simeq k^5$.
 - (d) The matrix $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ is in rational canonical form.
 - (e) $\mathbb{Z}/n\mathbb{Z}$ is a PID.
- (2) Show that we have the following natural isomorphisms of rings:
 - (a) $\text{End}(\mathbb{Z}/n\mathbb{Z}) \simeq \mathbb{Z}/n\mathbb{Z}$ for all $n \in \mathbb{Z}$;
 - (b) $\text{End}(\mathbb{Z}^r) \simeq M_{r \times r}(\mathbb{Z})$ where the right hand side is equipped with usual matrix addition and multiplication.
- (3) Use the existence of the rational canonical form to prove the following important fact: If $k \leq L$ is a field extension and $A, B \in M_n(k)$ are L -conjugate, then they are in fact k -conjugate.

For any $A \in M_n(k)$, denote the k -pair (k^n, T_A) by V_A . For any monic non-constant polynomial $q(x)$, write V_q for the k -pair $(k[x]/(q(x)), T)$, where T is given by multiplication-by- x .

Definition 2. A matrix $A \in M_n(k)$ is **diagonalizable** if it is k -conjugate to a diagonal matrix.

- (4) Show that the following are equivalent for a matrix $A \in M_n(k)$:
 - (a) A is diagonalizable over k .
 - (b) The k -pair V_A is isomorphic to $V_{x-\lambda_1} \times \cdots \times V_{x-\lambda_n}$ for some $\lambda_1, \dots, \lambda_n \in k$.

If $\lambda \in k$ and $n \in \mathbb{Z}_{\geq 1}$, the **Jordan block matrix** $J_\lambda(n) \in M_n(k)$ is the matrix

$$J_\lambda(n) = \begin{bmatrix} \lambda & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 1 & \cdots & 0 & 0 \\ 0 & 0 & \lambda & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{bmatrix} \in M_n(k)$$

with λ s along the diagonal, and 1s just above the diagonal.

- (5) Suppose that $q(x) = (x - \lambda)^n \in k[x]$. Show that the k -pair V_q is isomorphic to $V_{J_\lambda(n)}$.
Hint: You'll need to pick the right basis for V_q !

Definition 3. The **ring of Gaussian integers** $\mathbb{Z}[i]$ is the subset of $\mathbb{C}$ consisting of elements $a + bi$ with $a, b \in \mathbb{Z}$: This is in fact a sub-ring of $\mathbb{C}$.

- (6) Show that, for $\alpha, \beta \in \mathbb{Z}[i]$ with $\alpha \neq 0, \pm 1, \pm i$, there exist $\kappa, \rho \in \mathbb{Z}[i]$ with $|\rho| < |\alpha|$ such that $\beta = \kappa\alpha + \rho$. Use this to verify that $\mathbb{Z}[i]$ is a PID.¹

Hint: Looking at β/α , you just have to show that every element of $\mathbb{C}$ has distance less than 1 to the closest point with integer co-ordinates.

¹This is the complex absolute value here given by $|a + ib| = \sqrt{a^2 + b^2}$